

MISSION 100

JEE 2027

Lecture – 01

Physics

Mathematical Tools and Vector

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Physics Wallah





Topics to be Covered

1

Vector Basics

2

Resolution of Vectors

3

Vector Addition

4

Vector Multiplication



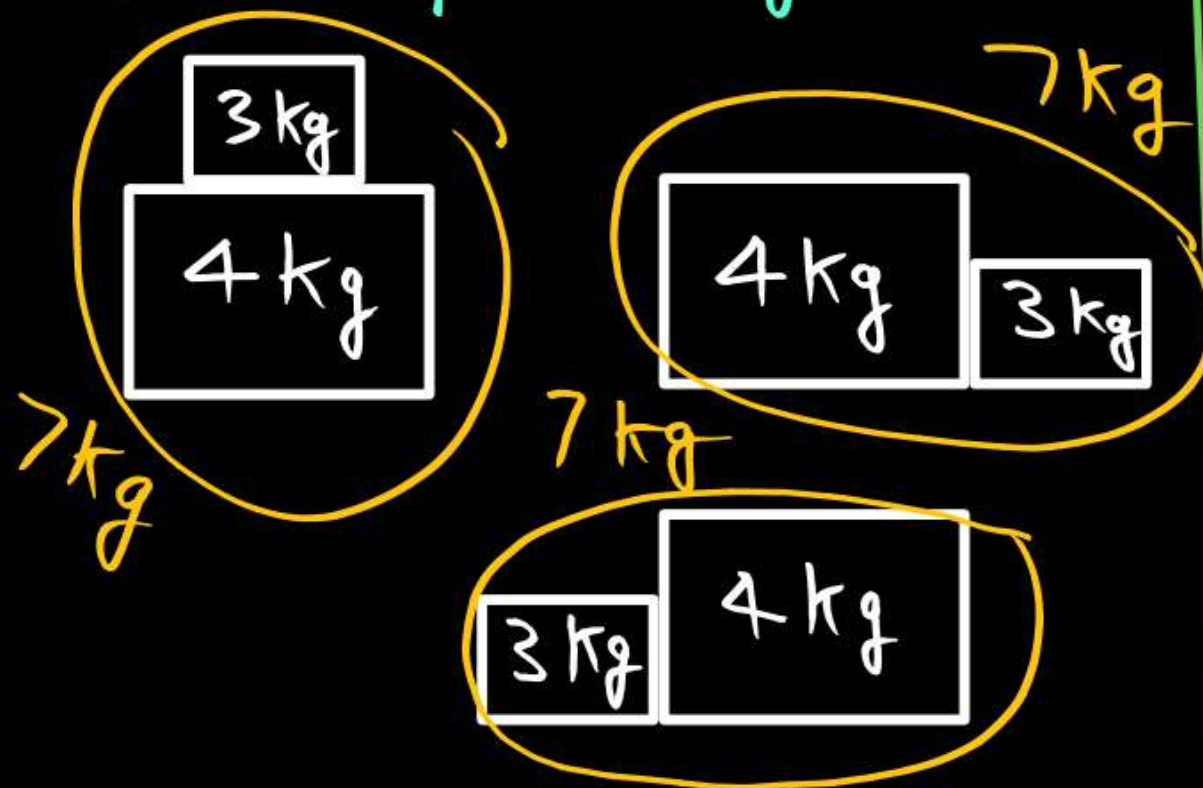
Scalar and Vector



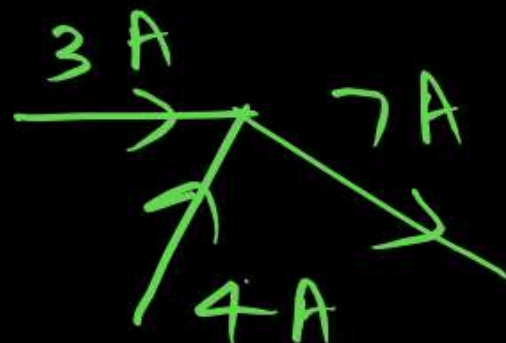
Physical Quantities

Scalar

- * only magnitude
- * Simple Algebra



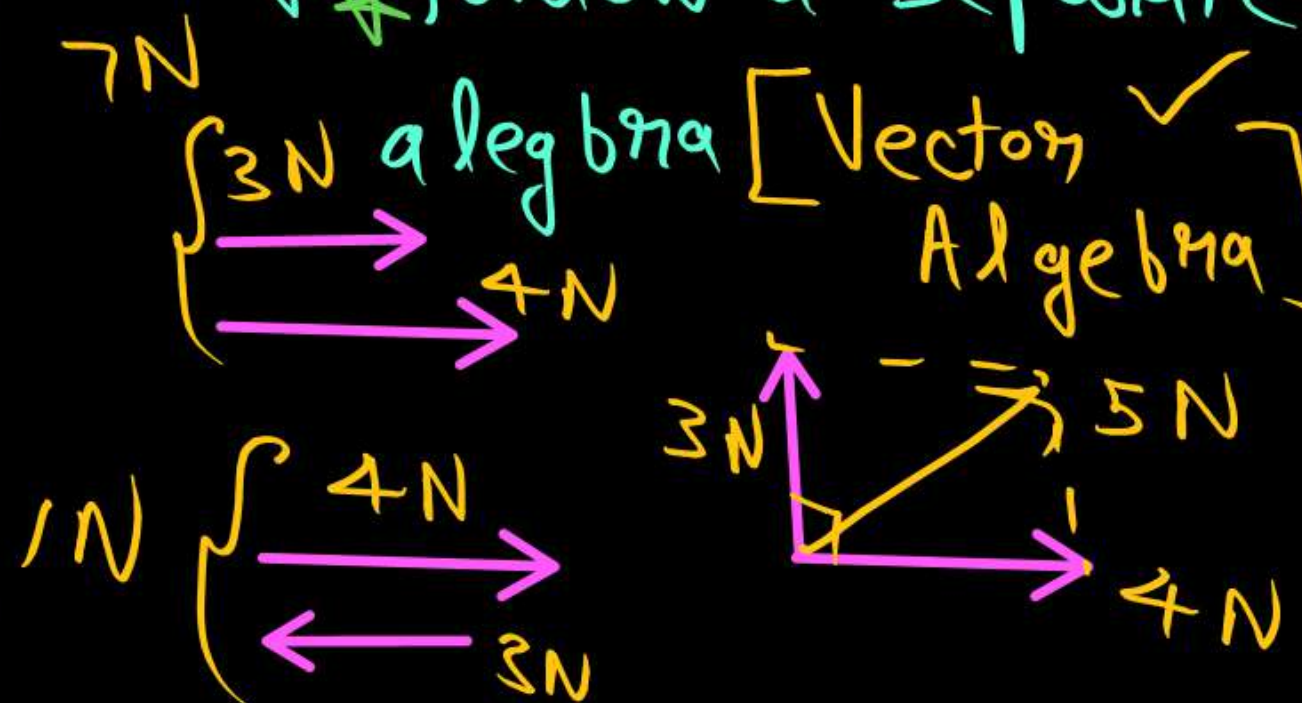
Current:



Vectors

- * both magnitude & dirⁿ

* follow a separate algebra [Vector Algebra]

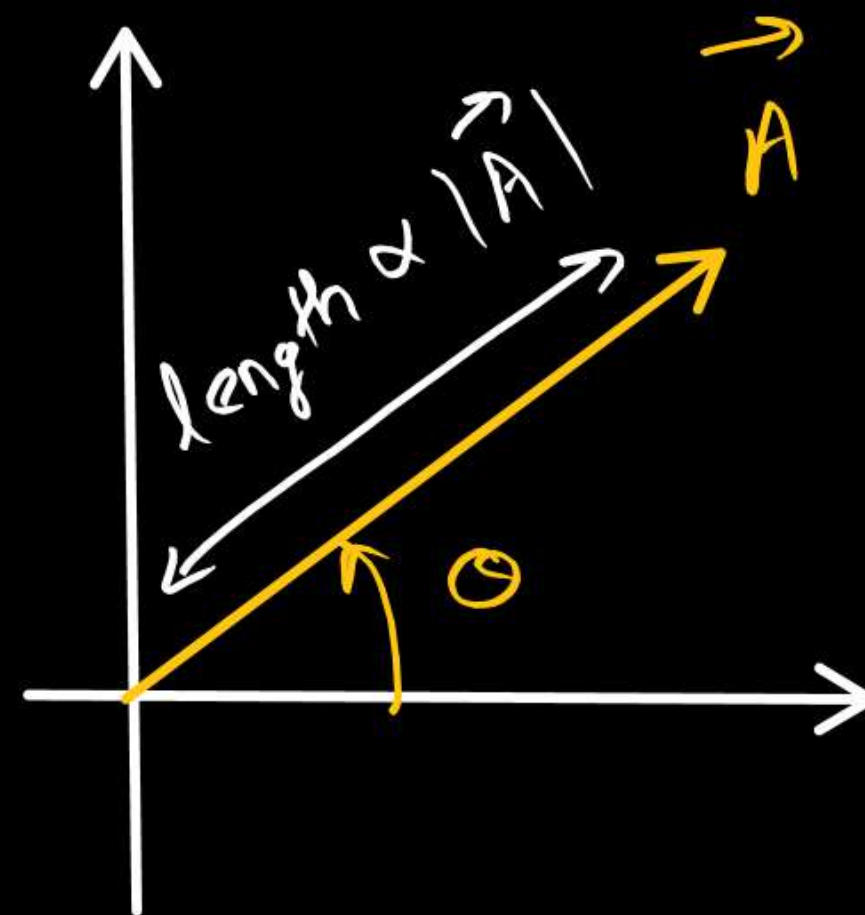




Representation

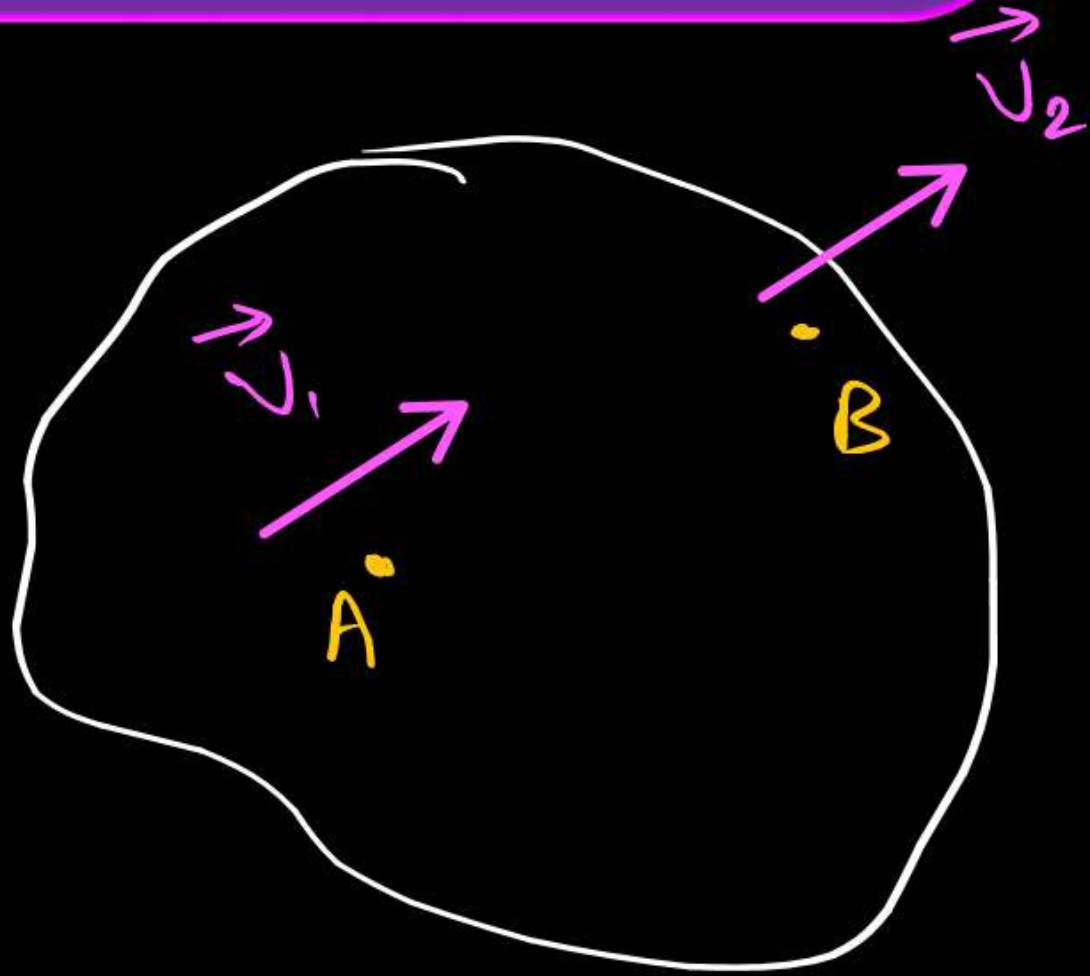
$\vec{A}$ $\rightarrow$ both magnitude
& dirⁿ

$|\vec{A}|$ = magnitude
of $\vec{A}$

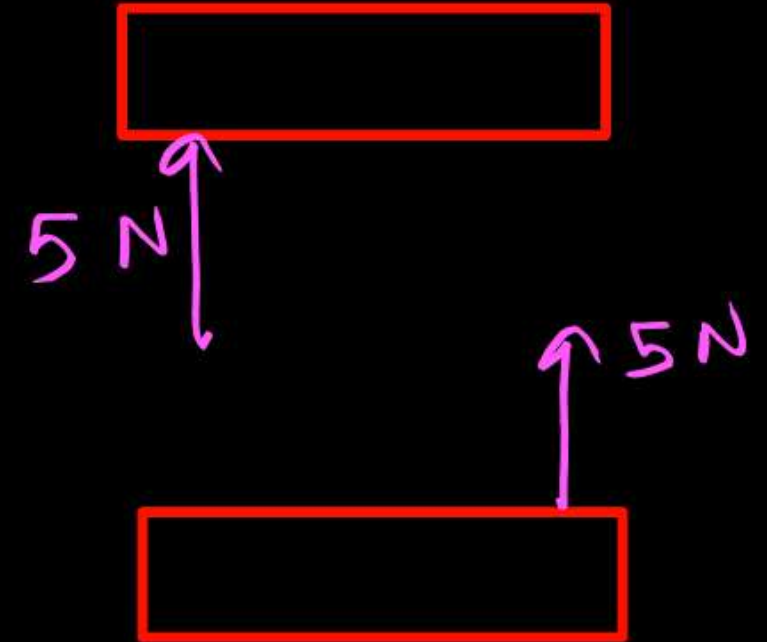




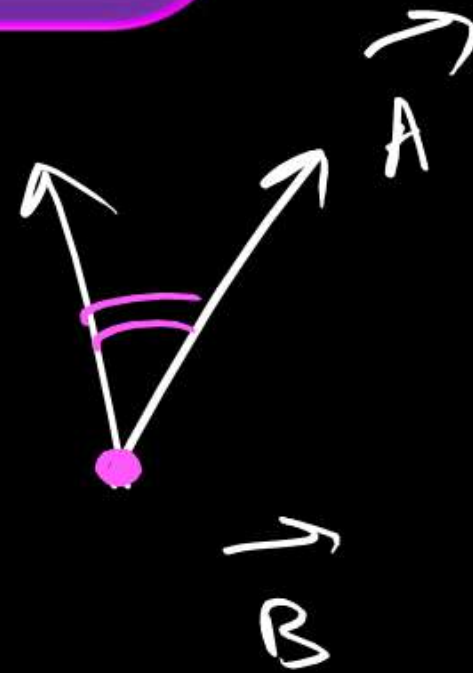
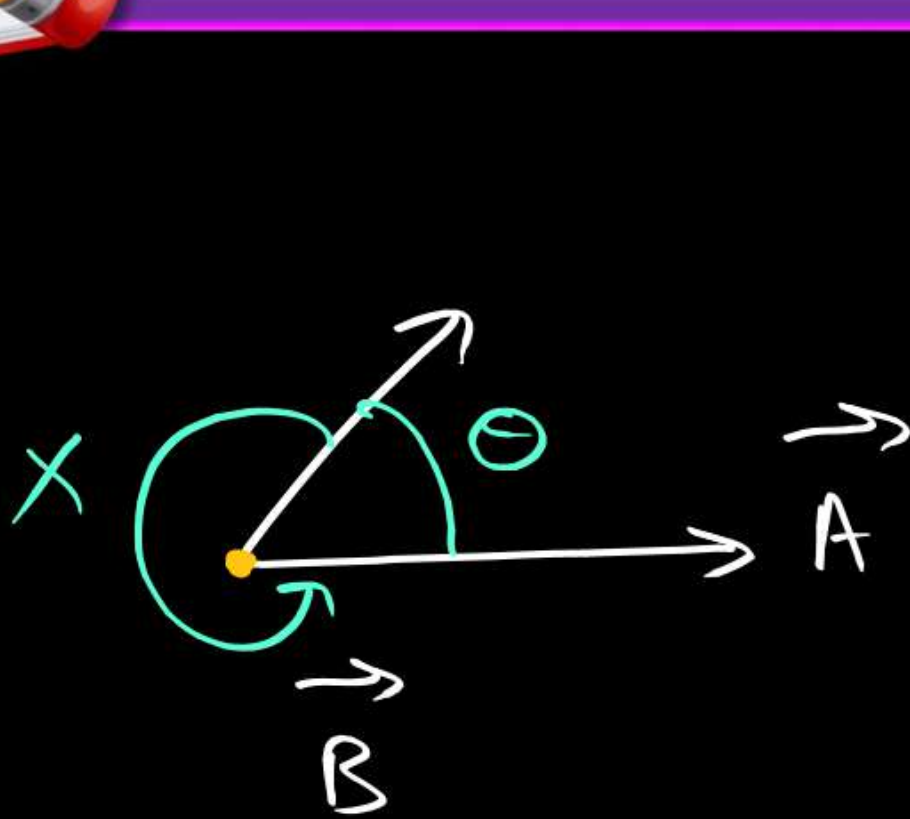
Parallel Transformation



$$\vec{v}_1 = \vec{v}_2$$

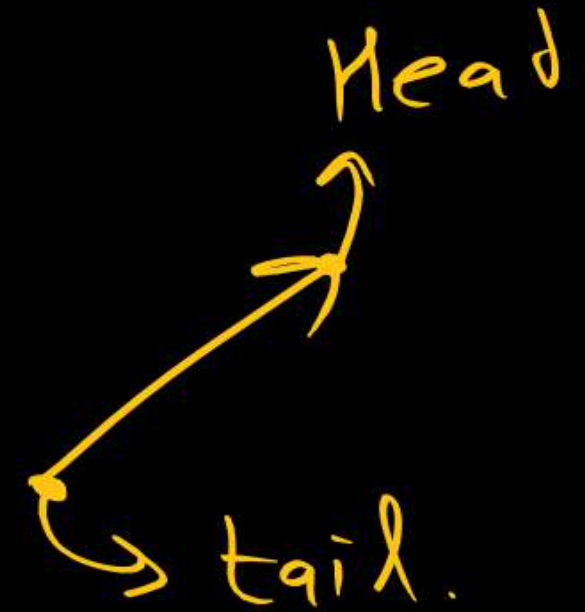


Angle Between Two Vectors



$$\theta \in [0^\circ, 180^\circ]$$

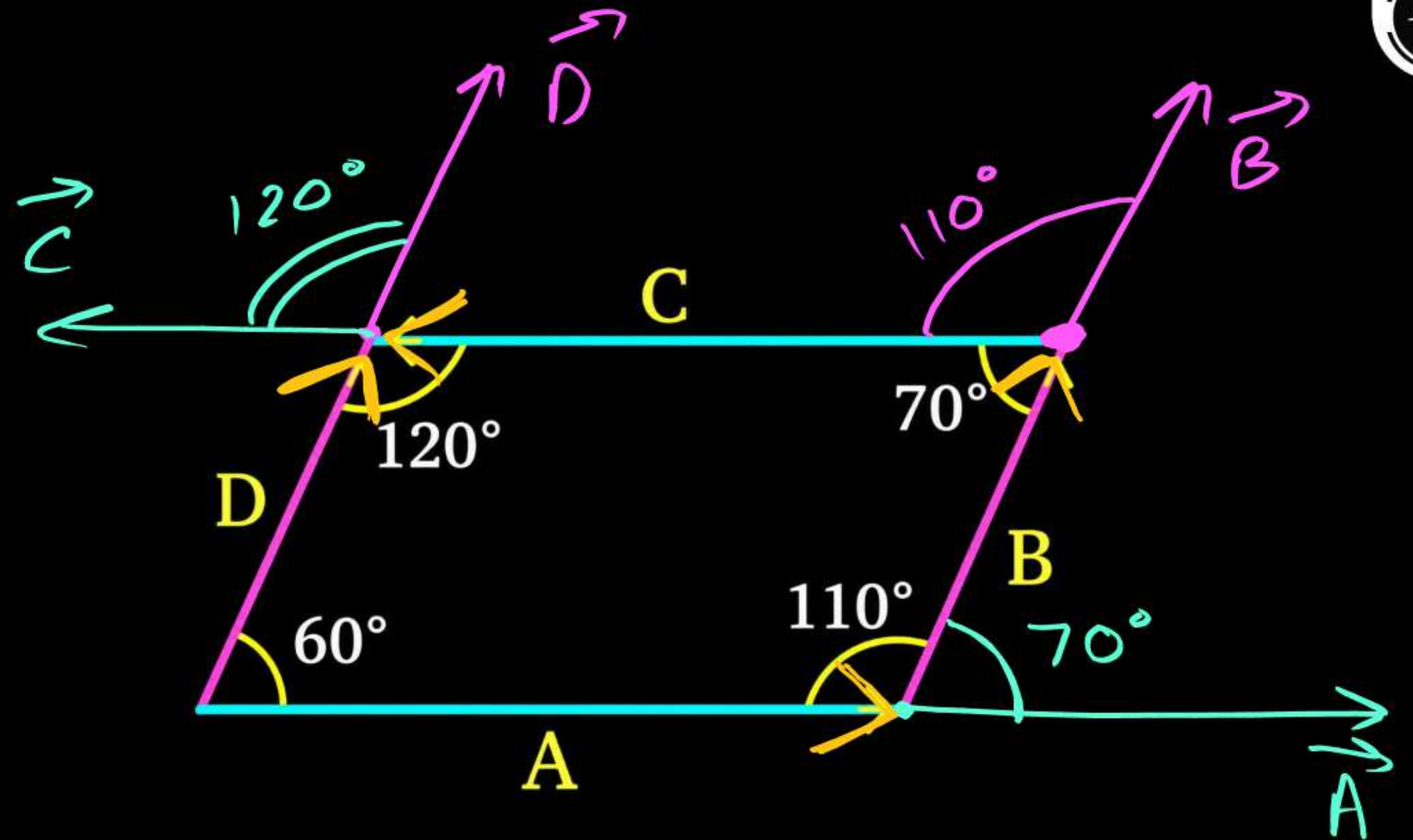
* Join the two
vectors tail
to tail.



Question

In the given figure

- ☒ 1 Angle between $\vec{A}$ and $\vec{B}$ is 110°
- ☒ 2 Angle between $\vec{C}$ and $\vec{D}$ is 60°
- ☒ 3 Angle between $\vec{B}$ and $\vec{C}$ is 110°
- ☐ 4 Angle between $\vec{B}$ and $\vec{C}$ is 70°





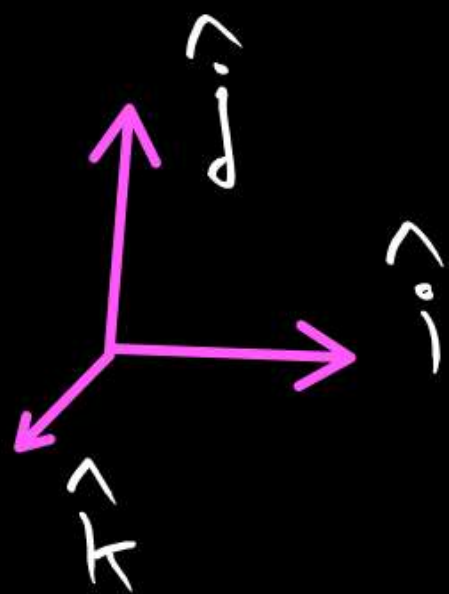
Unit Vector

→ dir^n batana

Representation: $\hat{A}$



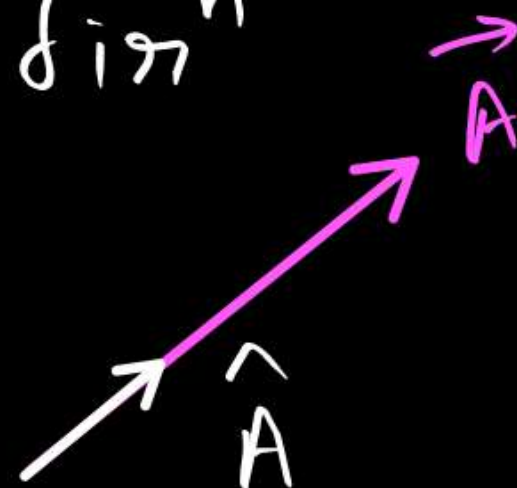
★ It is a vector having magnitude = 1 | $|\hat{A}| = 1$.
& points in a particular dir^n



$$\vec{v} = (5 \hat{i}) \text{ m/s.}$$

magnitude dir^n

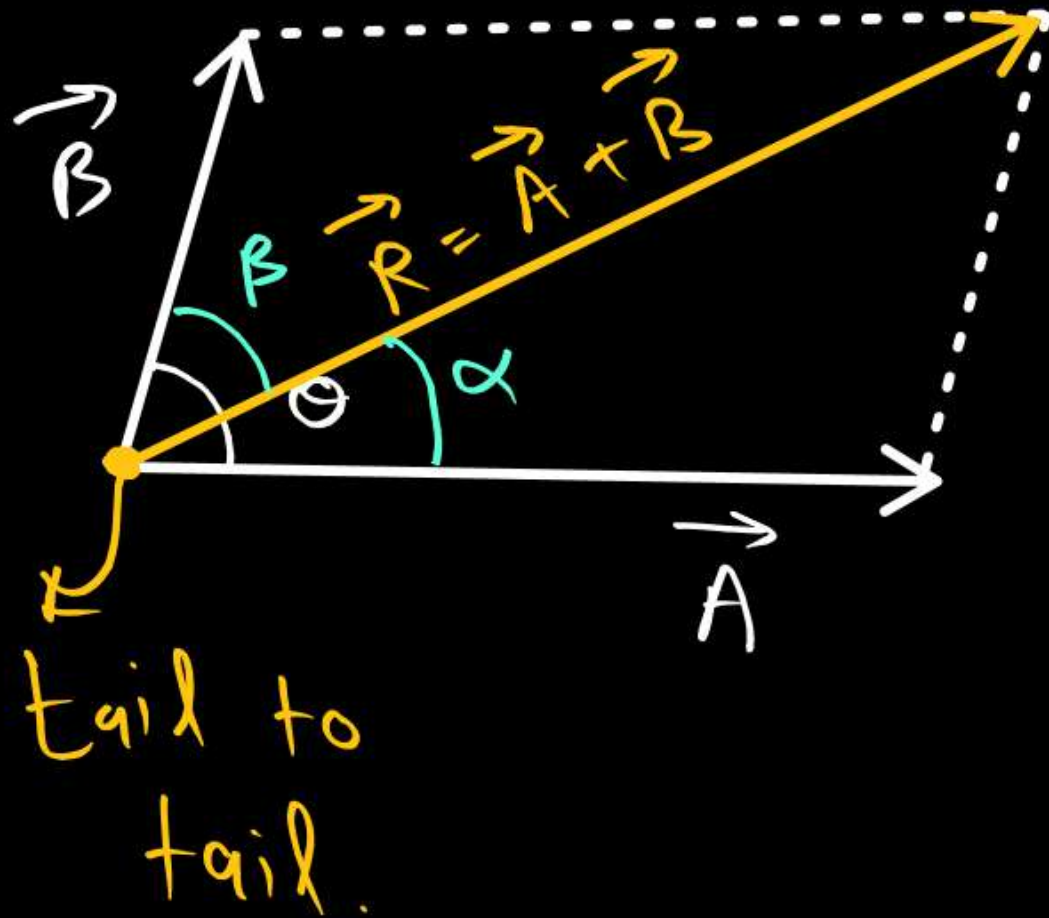
$$\hat{i} = \frac{8 \hat{i}}{8}$$



$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

$$\vec{A} = \underbrace{|\vec{A}|}_{\text{magnitude}} \underbrace{\hat{A}}_{\text{dir}^n}$$

Parallelogram Law of Vector Addition

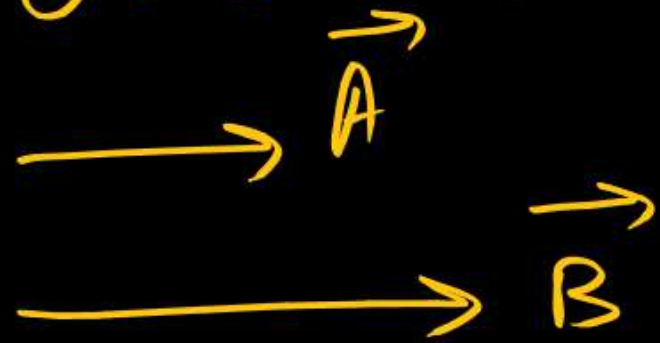


$$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

$$\tan \beta = \frac{A \sin \theta}{B + A \cos \theta}$$

① $\theta = 0^\circ$ [$\cos\theta \rightarrow \max$]



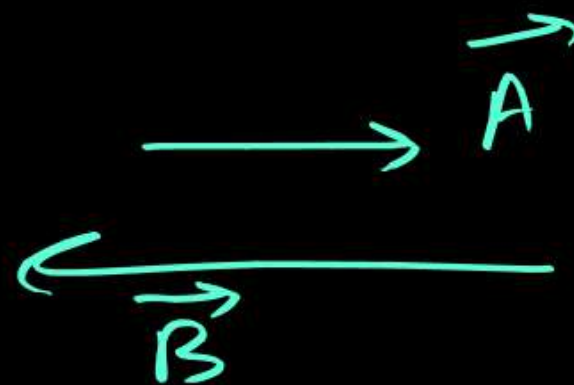
$$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB(1)}$$

$$= \sqrt{(A+B)^2}$$

$$\Rightarrow |\vec{R}|_{\max} = A + B$$

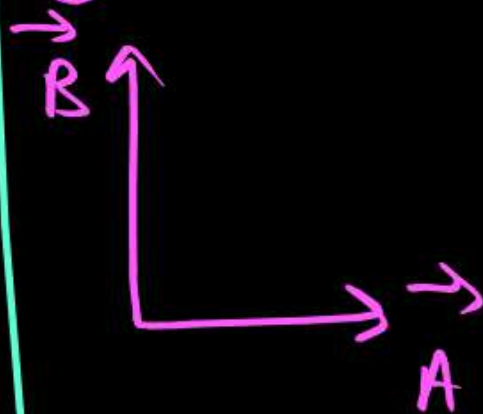
$$= |\vec{A}| + |\vec{B}|$$

② $\theta = 180^\circ$ [$\cos\theta \rightarrow \min$]



$$|\vec{R}|_{\min} = ||\vec{A}| - |\vec{B}||$$

③ $\theta = 90^\circ$



$$|\vec{R}| = \sqrt{A^2 + B^2}$$

$$||\vec{A}| - |\vec{B}|| \leq |\vec{R}| \leq |\vec{A}| + |\vec{B}|$$

$$||\vec{A}| - |\vec{B}|| \leq |\vec{A} + \vec{B}| \leq |\vec{A}| + |\vec{B}|$$

Question



Two forces of 4 dyne and 3 dyne act upon a body. The resultant force on the body can only be

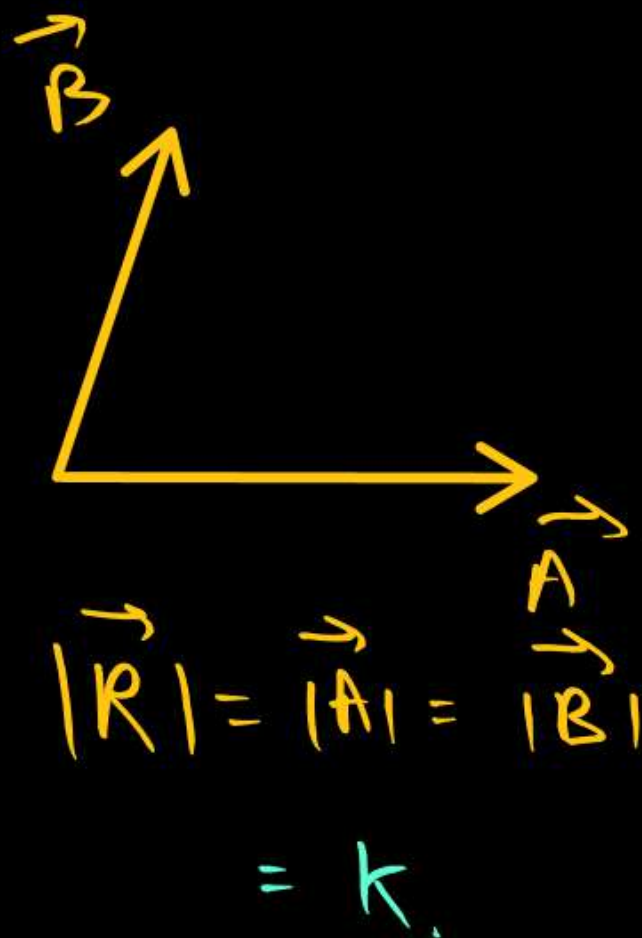
- ☐ 1 more than 3 dynes
- ☐ 2 more than 4 dynes
- ☐ 3 between 3 and 4 dynes
- ☒ 4 between 1 and 7 dynes

$$|\vec{A} - \vec{B}| \leq |\vec{R}| \leq |\vec{A} + \vec{B}|$$
$$1 \leq |\vec{R}| \leq 7.$$

Question

Two vectors have equal magnitudes. If their resultant also has the same magnitude, the angle between them will be:

- 1 30°
- 2 45°
- 3 120°
- 4 150°



$$K = \sqrt{K^2 + K^2 + 2(K)(K)\cos\theta}$$

$$K^2 = 2K^2 + 2K^2\cos\theta$$

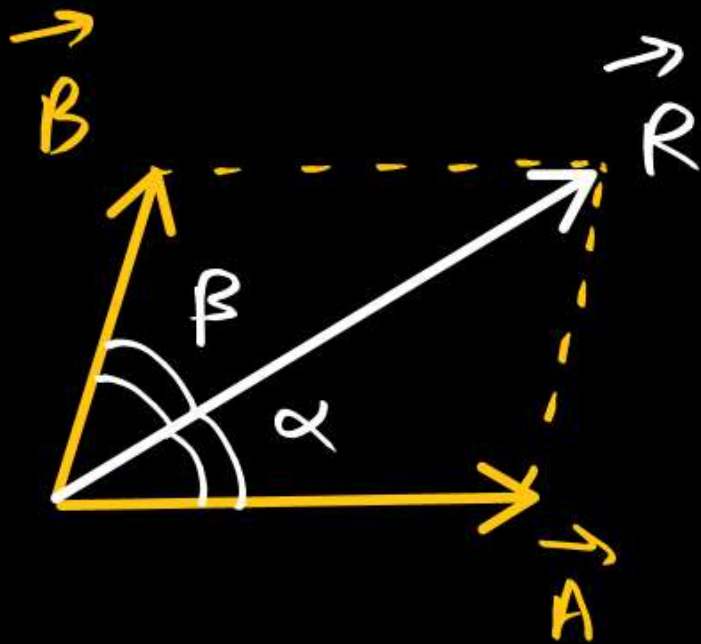
$$-K^2 = 2K^2\cos\theta$$

$$\Rightarrow \boxed{\cos\theta = -\frac{1}{2}}$$

$$\begin{aligned}\theta &= 180^\circ - 60^\circ \\ &= 120^\circ\end{aligned}$$

Question

Prove that if two vectors have equal magnitudes, then their resultant lies along their angle bisector.



$$\alpha = \beta = \frac{\theta}{2}$$

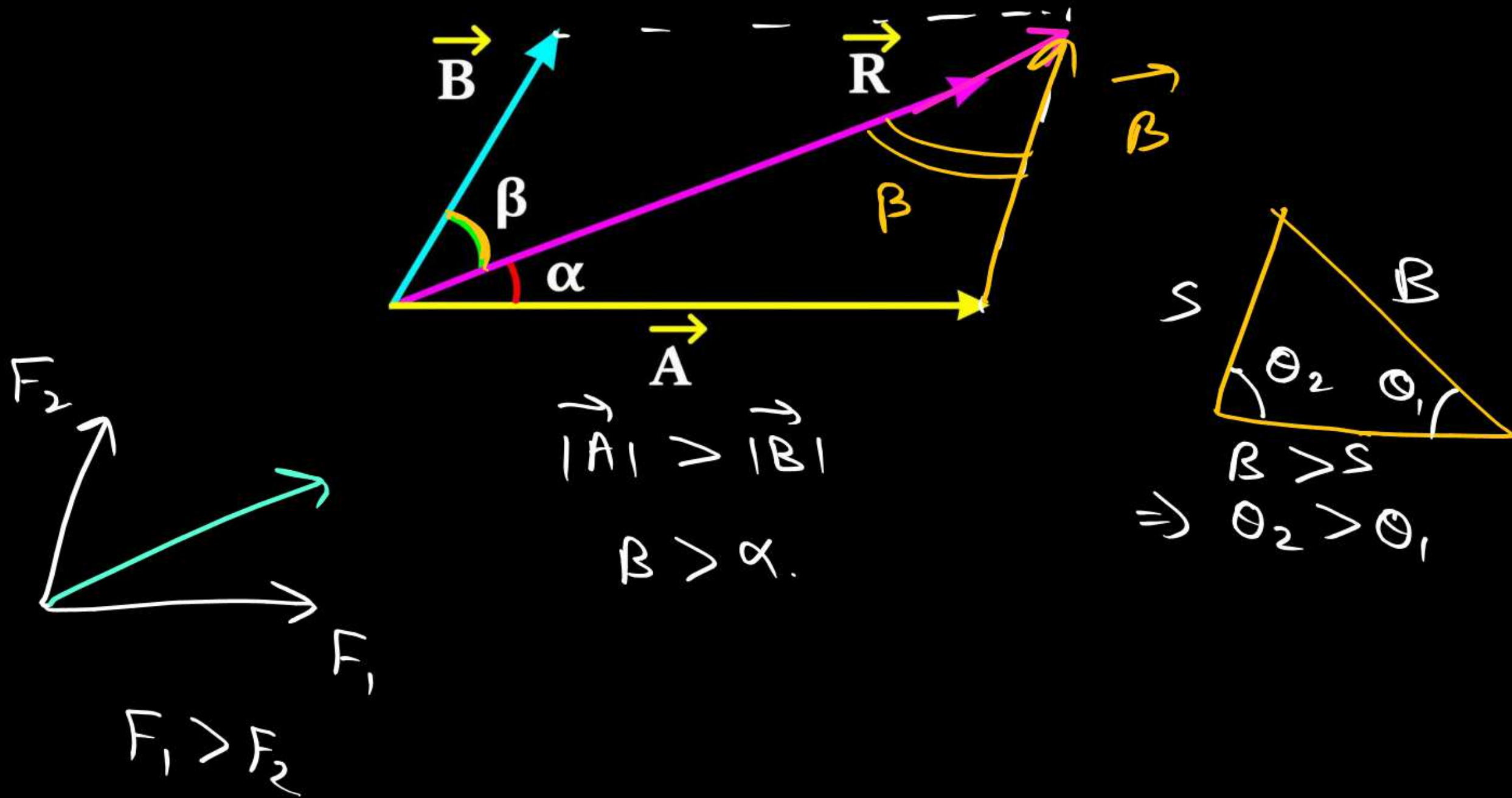
$$|\vec{A}| = |\vec{B}|$$

// $\square \rightarrow$ Rhombus.

Question

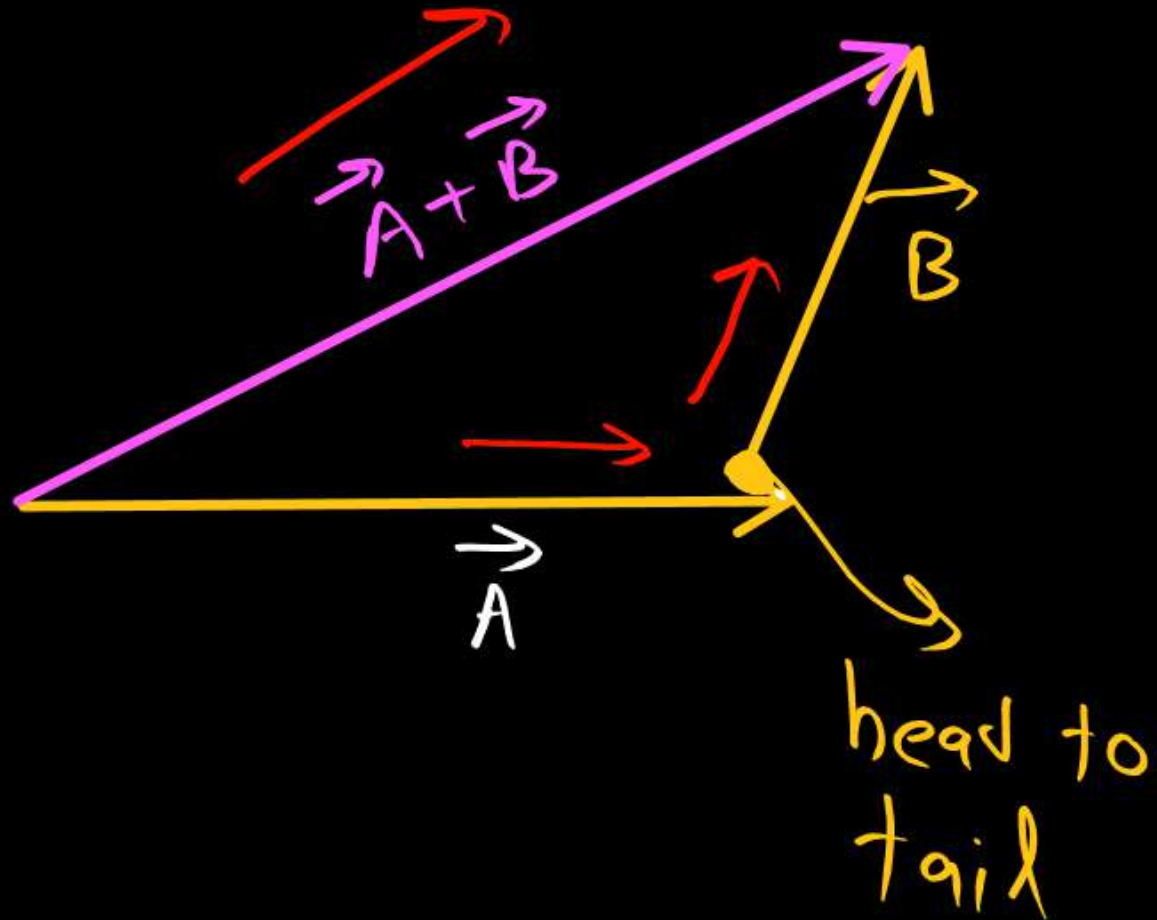
If $|\vec{A}| > |\vec{B}|$ then

- 1 $\alpha > \beta$
- 2 $\alpha < \beta$
- 3 $\alpha = \beta$
- 4 Can't predict





Triangle Law of Vector Addition

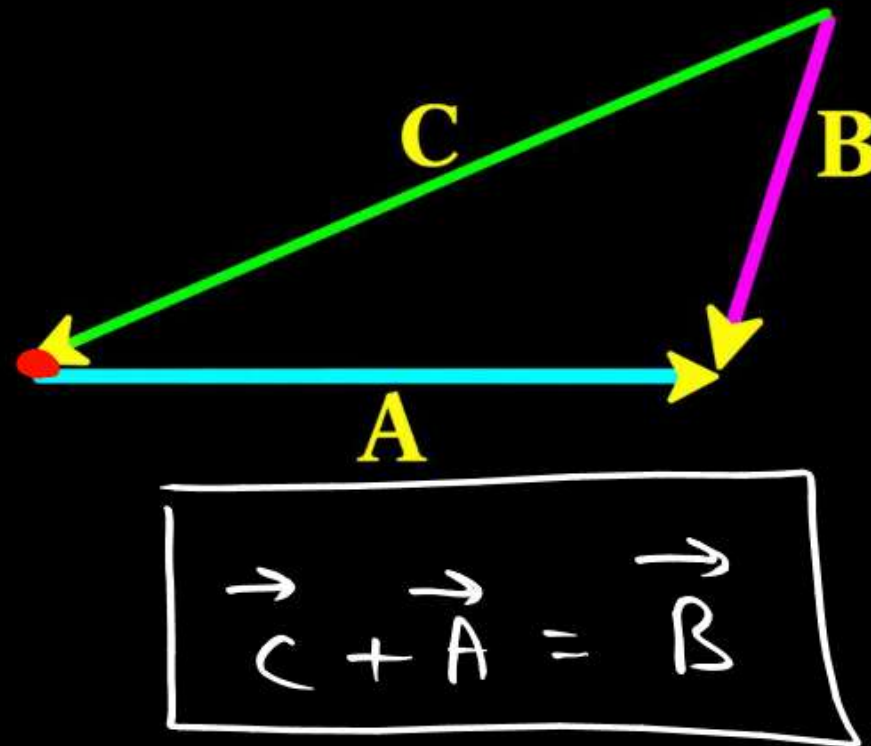


If two vectors are represented as two adjacent sides of $\triangle$ joined "head to tail" then their resultant is given by the 3rd side in opposite cyclic sense.

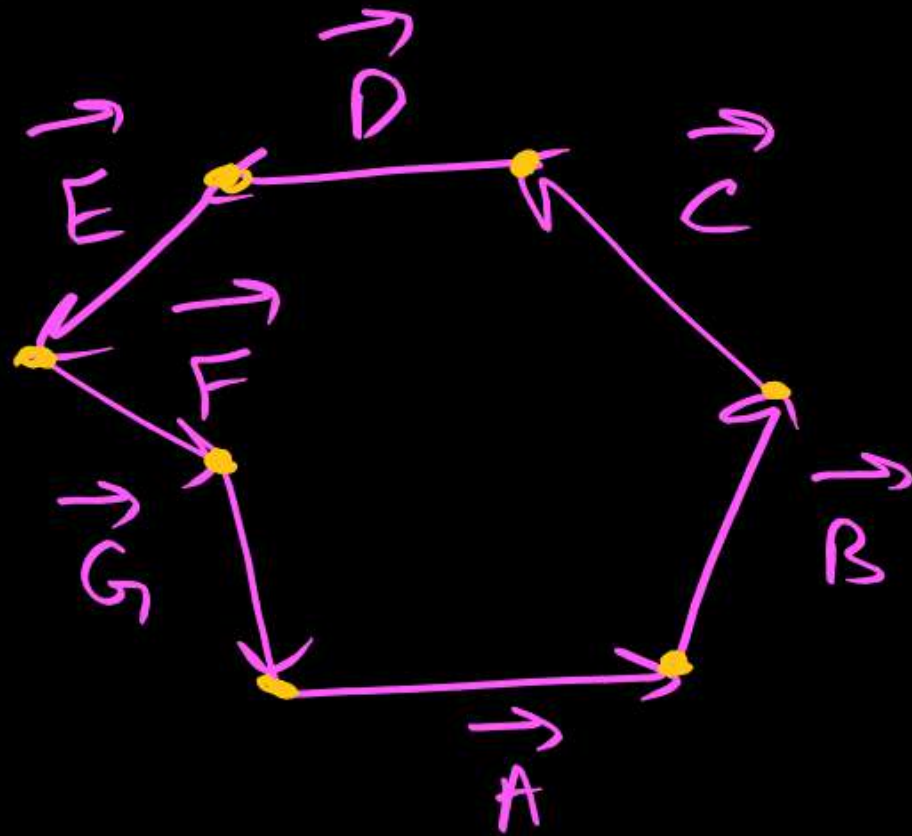
Question

For the figure

- 1 $A + B = C$
- 2 $B + C = A$
- 3 $C + A = B$
- 4 $A + B + C = 0$

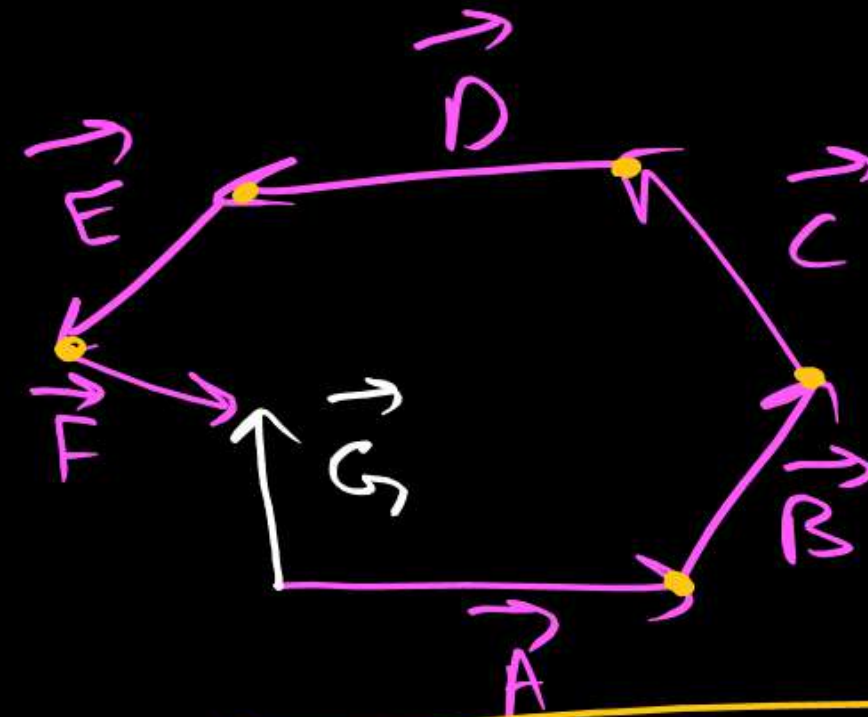


Polygon Law of Vector Addition



$$\vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E} + \vec{F} + \vec{G} = \vec{0}$$

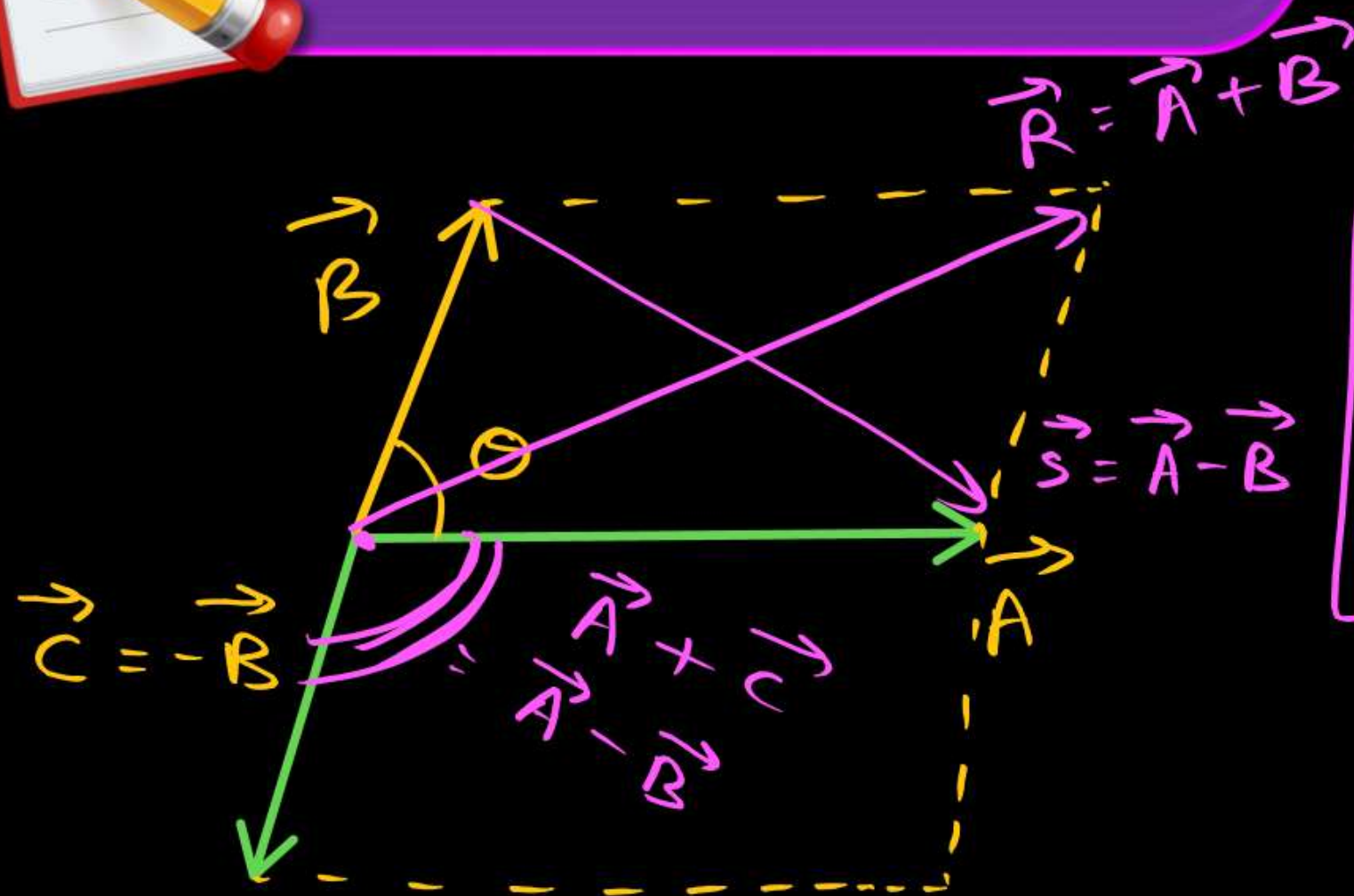
(all joined H to T)



$$\vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E} + \vec{F} = \vec{G}$$

Subtraction of Vectors

$$\vec{S} = \vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

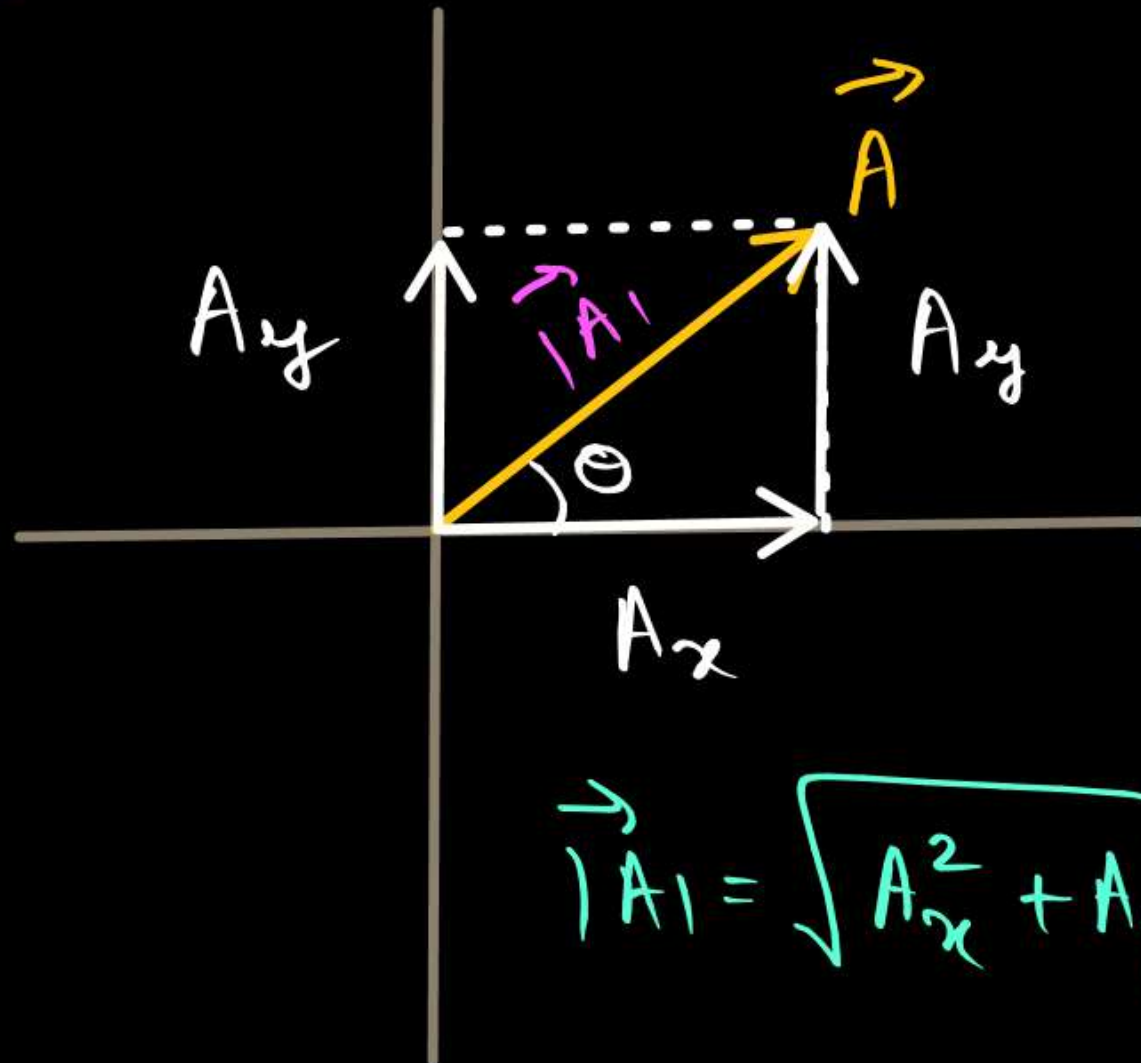


$$|\vec{S}| = \sqrt{A^2 + B^2 + 2AB \cos(180 - \theta)}$$

$$= \sqrt{A^2 + B^2 - 2AB \cos \theta}$$

θ = Angle b/w
 $\vec{A}$ & $\vec{B}$

Rectangular Components of Vector



$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

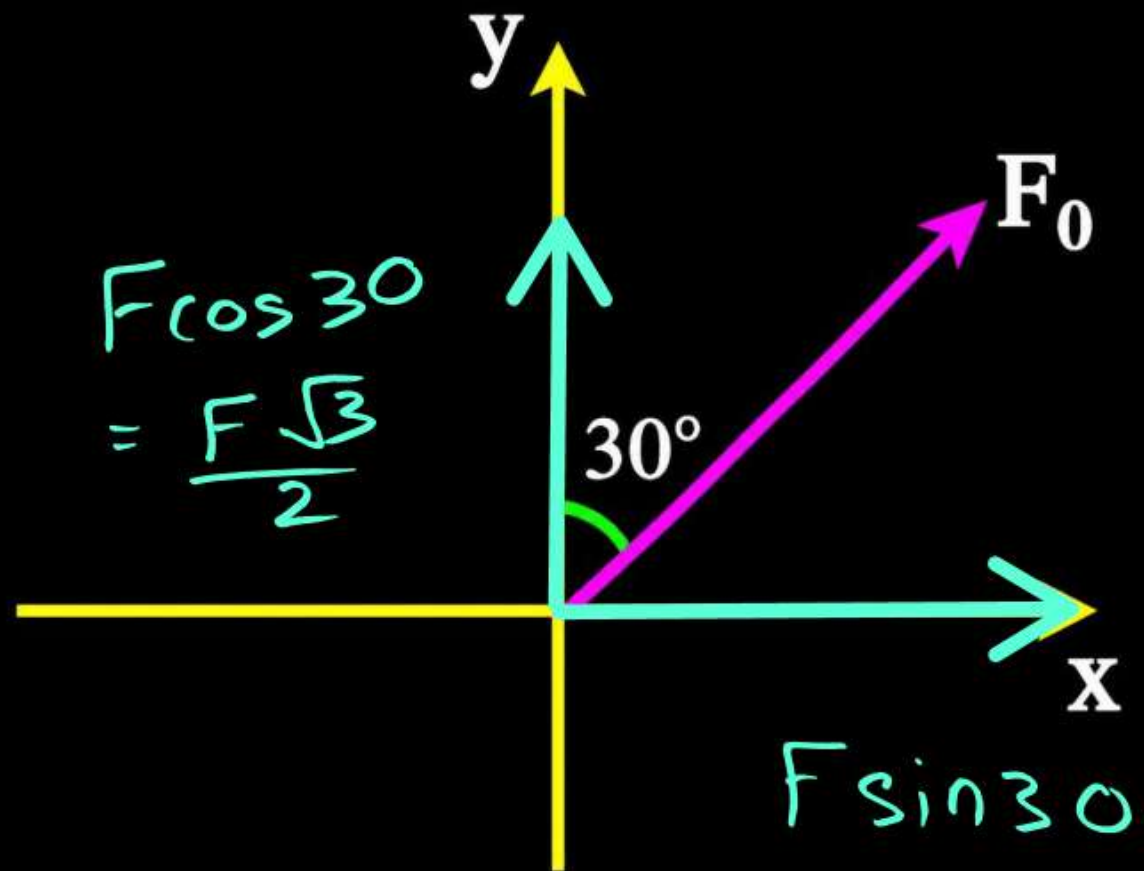
$$\frac{A_y}{|\vec{A}|} = \sin \theta \Rightarrow A_y = |\vec{A}| \sin \theta$$

$$\frac{A_x}{|\vec{A}|} = \cos \theta \Rightarrow A_x = |\vec{A}| \cos \theta$$

~~$x \rightarrow \cos \theta$
 $y \rightarrow \sin \theta$~~

θ se chipak $\rightarrow \cos \theta$
 Doosra wala $\rightarrow \sin \theta$

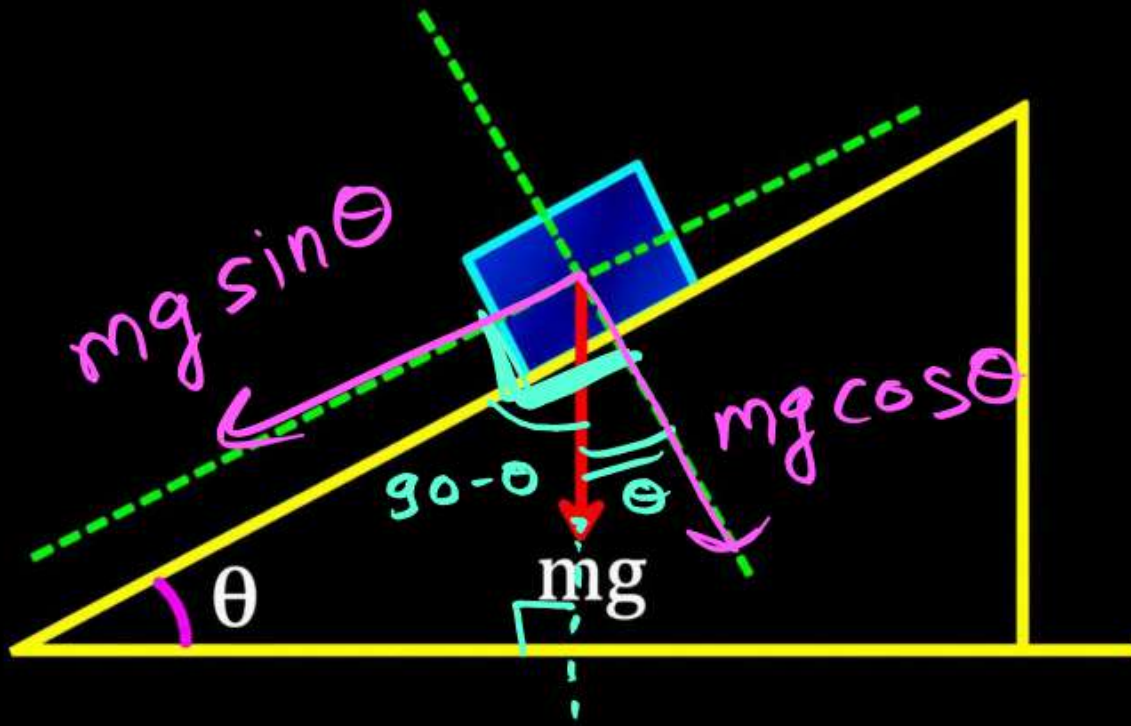
Question



$$x = \frac{F}{2}$$

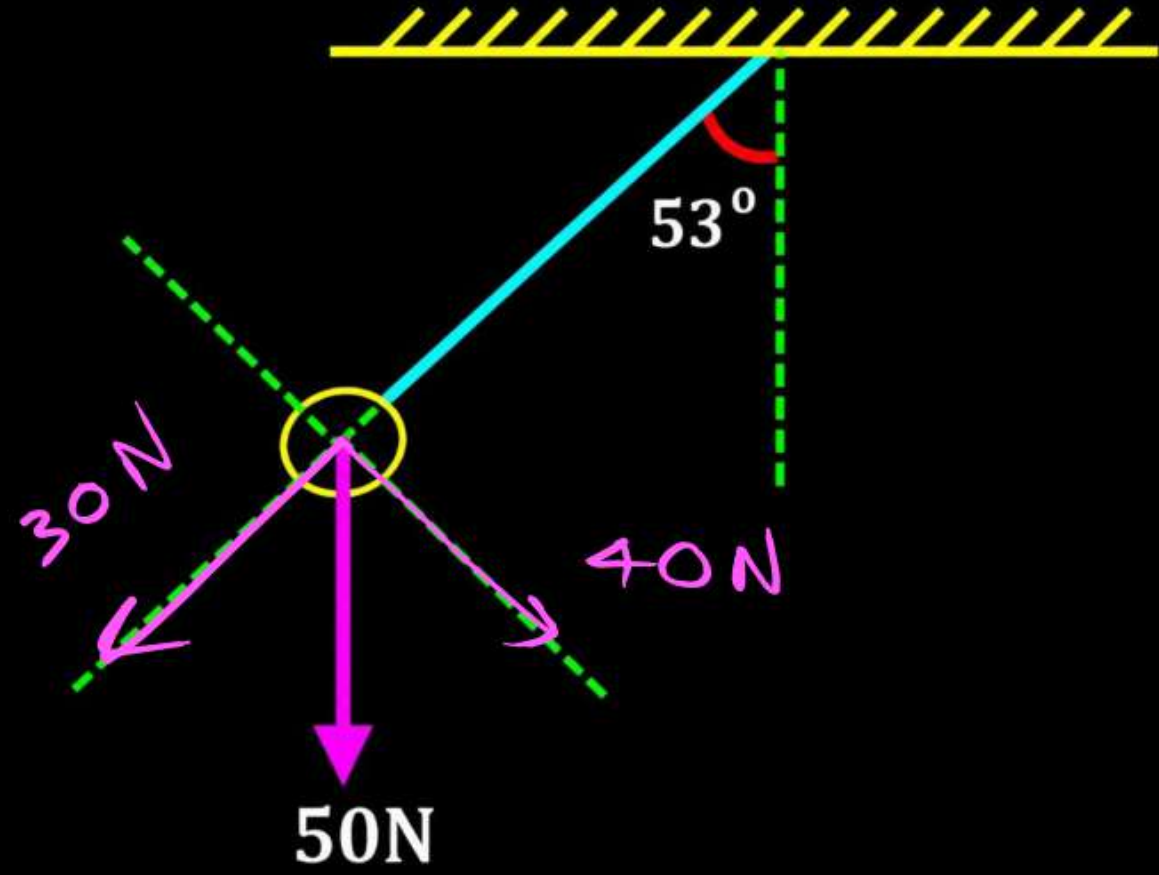
$$y = \frac{F\sqrt{3}}{2}$$

Question



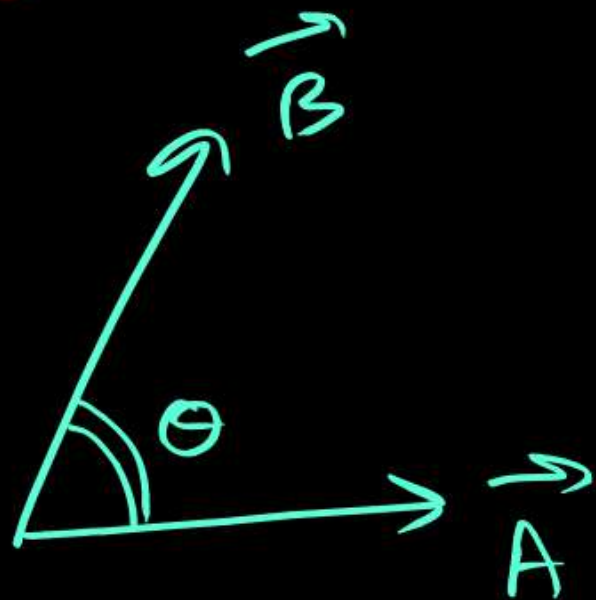
Question

→ H.W.



Dot Product of Vectors

→ Scalar Product



$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

↓
Scalar
Quantity.

$$\textcircled{1} \theta = 0^\circ \Rightarrow \vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}|$$

$$\textcircled{2} \theta = 180^\circ \Rightarrow \vec{A} \cdot \vec{B} = -|\vec{A}| |\vec{B}|$$

$$\textcircled{3} \theta = 90^\circ \Rightarrow \vec{A} \cdot \vec{B} = 0$$

$$(\vec{A} \perp \vec{B}) \Rightarrow \textcircled{5} \boxed{\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0}$$

$$\textcircled{4} \hat{i} \cdot \hat{i} = |\hat{i}| |\hat{i}| \cos 0 = 1 \times 1 \times 1 = 1$$

$$\boxed{\begin{aligned} \hat{i} \cdot \hat{i} \quad \hat{j} \cdot \hat{j} &= \hat{k} \cdot \hat{k} \\ &= 1 \end{aligned}}$$

$$\star \vec{A} = \underline{a_1} \hat{i} + \underline{a_2} \hat{j} + \underline{a_3} \hat{k}.$$

$$\vec{B} = \underline{b_1} \hat{i} + \underline{b_2} \hat{j} + \underline{b_3} \hat{k}.$$

$$\star \vec{A} \cdot \vec{B} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$\star$



Cross Product of Vectors



Question



The angle between the two vectors $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ will be

Question



If $\vec{A} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 4\hat{k}$ then value of $|\vec{A} \times \vec{B}|$ will be

Question



If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + \alpha\hat{k}$ is zero. Then, value of α will be _____.

[JEE Main-2022]



Summary



Homework

★ DPP solve

★ Addition & Components. (Only S.C.Q.)

THANK YOU

